

Claire Lee

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EDUCATION

Carnegie Mellon University

August 2023 - May 2026 (Expected)

B.S. in Electrical & Computer Engineering

GPA: 3.8/4.0

- **Relevant Coursework:** Computer Architecture, Computer Systems, Logic Design & Verification, Embedded Systems, Digital System Design, Digital Signal Processing, IC Design, Data Structures & Algorithms, Machine Learning
- **Organizations:** IEEE (2024-2026 Executive Board Chair), Women in ECE, Society of Women Engineers

TECHNICAL SKILLS

Programming Languages: C, SystemVerilog, Java, Python, R, HTML5/CSS, Assembly (x86, RISC-V RV32I)

Software Tools: Xilinx Vivado, VCS, SCADE, Arduino, SQL, MATLAB, Git, Linux, Bash, SVN, GDB, VIM, ArcGIS

Tools: Logic Analyzer, Oscilloscope, Signal Analyzer, 3D Printing, Soldering, Cadence Virtuoso

WORK EXPERIENCE

General Atomics

San Diego, CA

Autopilot Development Intern

May 2025 - August 2025

- Developed 10+ SCADE operators across 3 software branches to automate MQ-9B UAV area navigation, including takeoff & landing and holding pattern loiter through parameter computation, coordinate generation, and state machine integration
- Conducted 5+ regression tests using ATT framework in order to validate autopilot logic, testing 2,000+ requirements
- Created visualization tool for checking for cartographic distortion toward the North Pole through GoogleEarthPro and Python

CMU CyLab & CORGi Lab

Pittsburgh, PA

Research Assistant under Professor Vyas Sekar & Professor Nathan Beckmann

May 2024 - Present

- Developed full-stack tool (Node.js + React.js) to run and visualize results of 500+ cyberattack simulations
- Implemented custom cache in C for IPC-flattening scheduler, improving throughput and reducing data bottlenecks by 34%
- Automated trace evaluations that simulate cache reuse transformations in order to visualize memory access locality

FedEx

Pittsburgh, PA

Route Engineering Intern

May 2024 - August 2024

- Created dashboard of 600+ dispatch stations to visualize delivery metrics, performance trends, and vehicle GPS reliability
- Automated analysis of key performance metrics in Python and ArcGIS including driver at-stop times, off-route time, and GPS outages, increasing the overall accuracy of predicted routes by 33%
- Attended ESRI UC Conference, integrating the newly introduced path snapping and GeoAI-assisted routing features

UCSD UJIMA Lab & San Diego Super Computer Center

San Diego, CA

Machine Learning Research Intern

March 2022 - July 2023

- Analyzed 50,000+ Tweets using Python, R, and NLP techniques (TF-IDF, N-grams, topic modeling, sentiment analysis)
- Led bot detection initiative with Twitter/Botometer APIs, managing team of 4 and flagging 12,000+ Tweets
- Deployed recommender chatbot using Rasa with AWS-hosted backend and React.js, enabling ranked search of 3,000+ products

PROJECT EXPERIENCE

RISC-V Processor

April 2025

- Built a synthesizable RV32IM RISC-V processor with a 6-stage pipeline and dual-ported SRAM cache in SystemVerilog
- Optimized performance via caching, prefetching, branch prediction, and data forwarding, reaching 255 MIPS and 2390 MIPS/W

Dynamic Memory Allocator

November 2024

- Implemented malloc, calloc, realloc in C reaching 76% memory utilization and 10.5K ops/sec under fragmentation and alignment constraints using best fit and segregated free list strategies with coalescing and block splitting

Simulated Cache

October 2024

- Developed simulator to model direct-mapped and set-associative cache behavior with LRU replacement policies reaching a cache hit rate of 94%
- Applied loop tiling & memory reordering to maximize cache hits in matrix transpose, improving performance by over 10x

Underwater Autonomous Vehicle

July 2022 - August 2022

- Built autonomous underwater vehicle using PID control, ArduPilot, and GPS to collect real-time environmental data around a preplanned path in a body of water
- Implemented 3D printing to build buoy with custom sensor stabilizers for vehicle and integrated it with IC Design
- Integrated custom LoRa + Arduino communication network between buoy, base station, and vehicle using C
- Applied EDA, linear regression, K-means clustering, and goodness of fit to 400+ data points using Tensorflow library

AWARDS

Awards: [AutumnnHacks Hackathon 1st Place](#), \$1000 ACA & APAPA Digital Gallery Project Grant, San Diego City Council Technology Project Honor, Dean's List x3 (Fall 2023, Spring 2024, Fall 2024), National Merit Commended Scholar